



A CRISP-DM Framework for Transactional Knowledge Discovery and Intelligent Decision Support Using FP-Growth

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Abstract: Transaction data increasingly serves as a strategic knowledge source in data-driven decision support systems. However, many organizations still use transaction records primarily for operational purposes and do not systematically transform them into actionable knowledge that supports managerial decision-making. This study aims to investigate a knowledge discovery framework based on the Cross-Industry Standard Process for Data Mining (CRISP-DM) and the Frequent Pattern Growth (FP-Growth) algorithm for extracting transactional knowledge and supporting intelligent decision-making. A dataset containing 978 sales transactions and 157 active products was analyzed through six CRISP-DM stages, including business understanding, data understanding, data preparation, modeling, evaluation, and deployment. The FP-Growth algorithm was implemented to identify frequent itemsets and generate association rules. Twenty-one experimental scenarios involving different support and lift thresholds were conducted to determine an appropriate parameter configuration. The results showed that a minimum support value of 0.06 combined with a lift value of 2 produced a balanced set of association rules with strong business relevance. Stable purchasing patterns and contextual association patterns were identified. The extracted rules were integrated into a management information system to support product bundling and cross-selling functions. The results indicate that FP-Growth-based frequent pattern mining can transform transaction records into operational transactional knowledge with practical business value. This study demonstrates that integrating a complete CRISP-DM process with knowledge extraction and system deployment provides an effective pathway for developing data-driven marketing intelligence and decision-support mechanisms. The proposed framework provides a practical foundation for intelligent marketing systems and offers insights into knowledge discovery applications in transactional environments.

Keywords: Transactional knowledge discovery; Intelligent decision support; Frequent Pattern Growth; Cross-Industry Standard Process for Data Mining; Association rule mining; Management information systems

1 Introduction

Digital transformation in the business sector has encouraged organizations to shift from intuition-based decision making to a data-driven decision making (DDDM) approach. This approach treats data as a decisive asset, processed into knowledge to support both operational and strategic decisions. Various studies argued that DDDM could improve organizational performance through operational optimization, understanding customer behavior, and more precise strategic planning [1, 2]. In the context of marketing, the use of sales transaction data is attracting considerable interest because this could uncover purchasing patterns that are not immediately apparent but are highly relevant to the development of promotional strategies, product placement, and loyalty program design [3]. The retail and distribution sector is one of the most data-rich domains but is often suboptimal in its analytical processes. Sales transaction data is generally stored as monthly reports without being further processed into knowledge conducive to decision making. A search of the literature revealed that data-driven decision support systems in the retail and distribution sectors substantially improved performance, particularly in inventory management, customer analytics,

pricing, and promotion planning [4, 5]. When transaction data is not systematically used, companies risk missing opportunities to understand customer shopping patterns, identify potentially profitable product combinations, and determine the most effective marketing strategies for specific segments.

In the field of computer science and information systems, data mining is the primary approach for uncovering hidden patterns in large datasets, including sales transaction data. One important branch of data mining is association rule mining, which aims to find intriguing relationships between items in a transaction set [6, 7]. This process is often implemented through market basket analysis, where co-occurrence patterns are identified to support decisions such as product layout, promotional package design, and product recommendations [8, 9]. As the volume and complexity of data increase, the need for efficient methods to explore frequent patterns becomes increasingly urgent.

Among various association rule mining algorithms, Frequent Pattern Growth (FP-Growth) occupies a pivotal position due to its ability to mine frequent item sets without explicitly generating candidates as in Apriori. FP-Growth uses the FP-Tree structure to perform data compression and pattern mining more efficiently, optimizing it for relatively large transaction datasets [8, 10]. Recent studies have elucidated that FP-Growth was generally accepted in market basket analysis for various domains, such as book sales, groceries, and other retail products; it has been proven capable of generating association rules that are relevant for promotional needs, layout arrangements, and product recommendations [8, 9, 11].

Recent developments in association rule mining have also emphasized the importance of evaluating rule quality through metrics such as support, confidence, and lift. These metrics help researchers and practitioners distinguish between rules that appear frequently and those that actually exhibit strong and meaningful business relationships [6, 12]. Recent works have even proposed different variants and adjustments to elevate the value of the rules, thus enabling more effective rule selection and reducing noise in the resulting rule set. On the other hand, FP-Growth remains one of the most widely adopted algorithms for association rule mining due to its ability to efficiently discover frequent item sets without candidate generation. Its effectiveness in extracting purchasing patterns from transactional data has led to its extensive application in market basket analysis, recommendation systems, and data-driven decision support, hence confirming its relevance as an established frequent pattern mining technique [13, 14]. The systematic operation and replicability of the pattern mining process could be guaranteed in prior studies as they adopted the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework. CRISP-DM provides a structured set of stages from business understanding to deployment and has proven flexible across various data science domains, encompassing business and marketing analytics [15]. The application of CRISP-DM helps connect business needs with technical processes at the data mining level, so that the analysis results do not stop at the algorithmic level but are truly integrated into the organization's decision-making process.

Although FP-Growth has been prevalent in retail contexts [8, 9, 11], existing literature often presented association rules as static analytical reports rather than integrated knowledge artifacts. Furthermore, many studies employed arbitrary threshold values for support and confidence without a systematic optimization process. This paper bridged these gaps by implementing an end-to-end CRISP-DM pipeline that incorporates a support-lift grid search to ensure parameter robustness. The primary novelty of this work lies in transforming frequent patterns into feasible transactional knowledge, which is directly integrated into a distribution management information system to drive bundling and cross-selling features. In response to these requirements, this study utilized the FP-Growth algorithm within the CRISP-DM methodological framework to extract recurring patterns from historical transaction data of distribution companies in Gorontalo Province. All data was processed through the stages of business understanding, data understanding, data preparation, modeling, evaluation, and deployment to produce stable association rules that could be plausibly explained in business terms. In addition, the modeling process was accompanied by experiments on minimum support and lift parameterization to obtain the most optimal configuration for generating significant frequent patterns. Bearing this objective, the primary focus of this article is to illustrate how frequent pattern mining could substantially contribute to building transactional insights for distribution companies, as well as how a structured data mining pipeline could transform operational data into a basis for more accurate and evidence-based marketing decisions.

2 Methodology

This study was designed with the use of data-driven knowledge discovery approach and the FP-Growth algorithm within the CRISP-DM framework. This framework was chosen because it provides a holistic, systematic, and replicable analytical flow for the entire transaction pattern mining process, from understanding business problems to interpreting patterns as implementable knowledge to support managerial decisions.

Overall, this research method consisted of six main stages within the CRISP-DM framework, namely (1) Business Understanding; (2) Data Understanding; (3) Data Preparation; (4) Modeling; (5) Evaluation; and (6) Deployment/Knowledge Interpretation. Figure 1 illustrates the complete flow of the research stages in the frequent pattern mining process with the FP-Growth algorithm. This framework offered a methodical and iterative data mining approach to extracting knowledge from transaction data. Each stage was interconnected, beginning with the

identification of business needs and data understanding, followed by the data preparation and modeling processes utilizing FP-Growth. When proceeding to the evaluation stage of the quality of association rules, the support, confidence, and lift metrics were taken into account, before the final results were implemented as transactional knowledge to support decision making in marketing. Details of each stage are explained in the following sections.

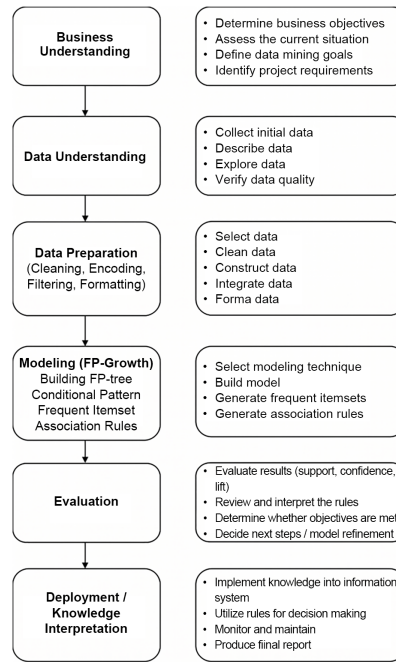


Figure 1. Cross-Industry Standard Process for Data Mining (CRISP-DM) stages

(1) Business Understanding

The initial stage focused on identifying analytical needs in the distribution company environment. The core problem that emerged was the company's inability to leverage sales transaction data as a source of strategic knowledge. All transactions were routinely recorded, but customer purchasing patterns, inter-product relationships, and repeated purchasing trends were not systematically analyzed, leading to marketing strategies that relied on intuition rather than empirical evidence. To deal with these challenges, this study set analytical objectives in the form of (a) extracting frequent item sets from sales transactions; (b) generating association rules based on support, confidence, and lift; and (c) interpreting these rules as transactional knowledge that could reinforce the formulation of marketing strategies such as product bundling and cross-selling.

(2) Data Understanding

The dataset consisted of 978 sales transactions recorded over a one-year operational period at a distribution company in Gorontalo Province. Each transaction included a list of purchased products, the purchase amount, and the transaction ID (invoice ID). The catalog of the company showcased 157 active products, in support of the formation of complex transaction combinations; therefore, this dataset was deemed relevant for frequent pattern mining.

Regarding the dataset scale, although 978 transactions might appear modest compared with high-frequency retail benchmarks, it was highly representative for a middle-scale distribution context. Unlike consumer retail, distribution data involved B2B-style transactions, in which each invoice represented bulk purchases with high item density, rendering it sufficient to extract stable and statistically significant association rules. Previous case studies demonstrated that a dataset in this range was appropriate for identifying meaningful regional supply chain patterns, when the FP-Growth algorithm was employed.

At this preliminary stage, an exploration was conducted to understand the dataset structure, product purchase distribution, item frequency, and data consistency. This initial analysis uncovered groups of products that consistently appeared together across transactions, indicating potential interpretation for significant association patterns. The dataset was then mapped into a transactional basket form, which was a set of unique products per transaction, as a standard form for the process of market basket analysis.

(3) Data Preparation

The data preparation stage ensured the dataset was in a format ready for processing by the FP-Growth algorithm. Based on minimum support, this process underwent data cleaning, transaction aggregation, encoding, and feature filtering. This stage produced a structured dataset ready for generation of frequent pattern.

(4) Modeling

Forming the core of this research, the modeling stage propelled the application of the FP-Growth algorithm to identify product patterns that frequently co-occurred (frequent item sets):

(a) FP-Tree Construction Process: FP-Growth constructed a transaction compression tree (FP-Tree), which accelerated pattern enumeration by eliminating the demand for candidate generation as in Apriori and enabling recursive traversal to generate conditional pattern bases. FP-Tree automatically captured relationships among products that frequently co-occurred.

(b) Frequent Item set Mining: Once the FP-Tree was formed, the algorithm performed a recursive process to unravel all item combinations that met the minimum support threshold. The support formula used is as follows:

$$\text{support}(A) = \frac{\text{number of transactions containing } A}{\text{total transactions}} \quad (1)$$

(c) Association Rule Generation:

$$\text{confidence}(A \rightarrow B) = \frac{\text{support}(A \cup B)}{\text{support}(A)} \quad (2)$$

$$\text{lift}(A \rightarrow B) = \frac{\text{confidence}(A \rightarrow B)}{\text{support}(B)} \quad (3)$$

Lift is the primary metric for assessing the strength of relationships between products, as it could determine causal relationships that would not occur randomly.

(d) Parameter Optimization: Support $\times$ Lift Grid Search

To determine the optimal parameters, this study engaged in 21 experimental scenarios combining support values (0.03–0.09) and lift values (1.5–3). The goal of optimization was to balance the number of rules and their relevance, in order to avoid overfitting (too low support) and underfitting (too high support). When interpreted with caution, the experimental results pointed out that the combination of support = 0.06 and lift = 2.0 produced rules with the most stable relevance and the lowest noise level.

(5) Evaluation

The evaluation helped ensure that the generated association rules met analytical and interpretative quality standards. The evaluation criteria included (a) pattern validity, which investigated whether the product combinations had business significance; (b) rule strength, which was measured based on confidence and lift. Rules with lift > 1.0 were considered statistically significant; and (c) pattern consistency, which ascertained whether the pattern emerged consistently across numerous transactions, not merely a small subset of the dataset. This evaluation was crucial as the association rules could be truly considered transactional knowledge.

(6) Deployment

The final stage was to translate the analysis results into practical knowledge. The patterns generated were mapped to tentative business strategies, such as product bundling recommendations, cross-selling strategies, product grouping in catalogs, and storage layout optimization. Although the study did not establish a complete system, these patterns were still interpreted by distribution companies in the operational context as a form of knowledge deployment.

3 Results

3.1 Overview of Transaction Data and Pre-processing Results

The data analyzed originated from the sales management information system of a distribution company in Gorontalo Province. The major dataset consisted of 978 sales transactions, with 157 active products recorded during the 2023 operational period. This historical dataset was used as the primary data source for the frequent pattern mining process, covering FP-Tree construction, frequent item set extraction, and association rules generation with the FP-Growth algorithm. Besides, 30 transactions were selected as a data subset to illustrate the FP-Tree construction process and facilitate visual interpretation of patterns.

During the deployment phase of the proposed framework, a web-based application was developed and implemented at the partner company. The transaction data exhibited in Table 1 is an example of operational data entered into application during the system implementation phase in 2025. This data was used to illustrate the functionality of the application, yet differed from the 2023 historical dataset used in the FP-Growth modeling and evaluation process.

In the data preparation stage of the CRISP-DM framework, the raw data was cleaned (duplicate removal, missing value handling, and product name standardization) and transformed into a market basket format, in which each transaction row was represented as a set of items (products) purchased together. This representation is displayed in Table 2.

Table 1. Sample of transactional record

No.	Invoice ID	Invoice Date	Product Purchased	Quantity
1	INV-01082025-0002	8/1/2025 0:00	Product 4 (Kapal Api)	4
			Product 16 (Silver Queen)	1
			Product 15 (Indomie)	7
			Product 10 (Wardah)	2
			Product 9 (Indomie)	3
2	INV-01092025-0001	9/1/2025 0:00	Product 18 (Wardah)	7
			Product 19 (Kapal Api)	6
			Product 17 (Kapal Api)	1
			Product 15 (Indomie)	1
			Product 4 (Kapal Api)	5
3	INV-01092025-0002	9/1/2025 0:00	Product 10 (Wardah)	4
			Product 13 (Silver Queen)	4
			Product 4 (Kapal Api)	3
			Product 13 (Silver Queen)	4
			Product 15 (Indomie)	6
			Product 10 (Wardah)	8
			Product 17 (Kapal Api)	8

Table 2. Sales transaction data transformation

No.	Transaction ID	Product Code
1	INV-01082025-0002	P04, P016, P015, P010
2	INV-01092025-0001	P09, P018, P019, P017, P015, P04, P010, P013
3	INV-01092025-0002	P04, P013, P015, P010, P017
4	INV-01092025-0003	P018, P02, P015, P04
5	INV-01092025-0004	P015, P010, P04, P013, P017
6	INV-02072025-0001	P08, P02, P018, P015, P04, P010
7	INV-02072025-0002	P016, P019, P010, P015, P04, P013
8	INV-02072025-0003	P09, P019, P015, P04, P010
9	INV-02082025-0001	P08, P06, P015, P04
10	INV-02092025-0001	P020, P02, P018, P015, P04, P010, P013
11	INV-02092025-0002	P015, P06, P04, P010
12	INV-02092025-0003	P03, P018, P019, P011, P015, P04, P010, P013
13	INV-02092025-0004	P03, P07, P018, P015, P04, P010
14	INV-02092025-0005	P015, P04, P010, P013
15	INV-02092025-0006	P02, P016, P015, P04, P010
16	INV-03072025-0001	P06, P012, P013, P017, P015, P04, P010, P07
17	INV-03082025-0001	P07, P019, P04, P015, P010, P013
18	INV-03082025-0002	P04, P015, P08, P010, P013, P017
19	INV-03082025-0003	P02, P08, P019, P07, P015, P04, P010, P013
20	INV-03092025-0001	P07, P018, P015, P04, P010
21	INV-03092025-0002	P019, P02, P018, P09, P015, P04, P010, P013
22	INV-03092025-0003	P017, P015, P013, P04, P010, P07, P016
23	INV-04082025-0001	P09, P018, P015, P04, P010
24	INV-04092025-0001	P05, P018, P015, P04
25	INV-04092025-0002	P016, P05, P04, P015, P010, P013
26	INV-05072025-0001	P016, P05, P07, P017, P015, P04, P010, P013
27	INV-05082025-0001	P017, P05, P015, P04
28	INV-05082025-0002	P016, P017, P012, P04, P015, P010, P013, P07
29	INV-05092025-0001	P011, P02, P019, P015, P04, P010, P013
30	INV-05092025-0002	P017, P06, P013, P015, P04, P010, P07

Next, the researcher calculated the frequency of each product's appearance in 30 sample transactions (as an initial illustration). Some products such as codes P04 and P015 emerged in all 30 transactions (100% support), while others emerged less frequently. This distribution pattern proved the existence of "core items" that were very dominant in a customer's shopping basket.

Methodologically, the data understanding and data preparation phases were important to guarantee that the frequent pattern mining process was applied to data that is consistent, structured, and ready for extraction into transactional knowledge. This aligns with best practices in other studies that applied the FP-Growth to purchase pattern analysis and marketing-strategy optimization in retail and Micro, Small and Medium Enterprises (MSMEs).

3.2 Frequent Pattern Mining Results and Frequent Pattern Tree Formation

After transforming the data into a basket format, the FP-Growth algorithm was employed to extract frequent item sets and build the FP-Tree. For illustrative purposes, the paper presented the FP-Tree construction process in stages using 30 sample transactions, starting with the first transaction (Figure 2).

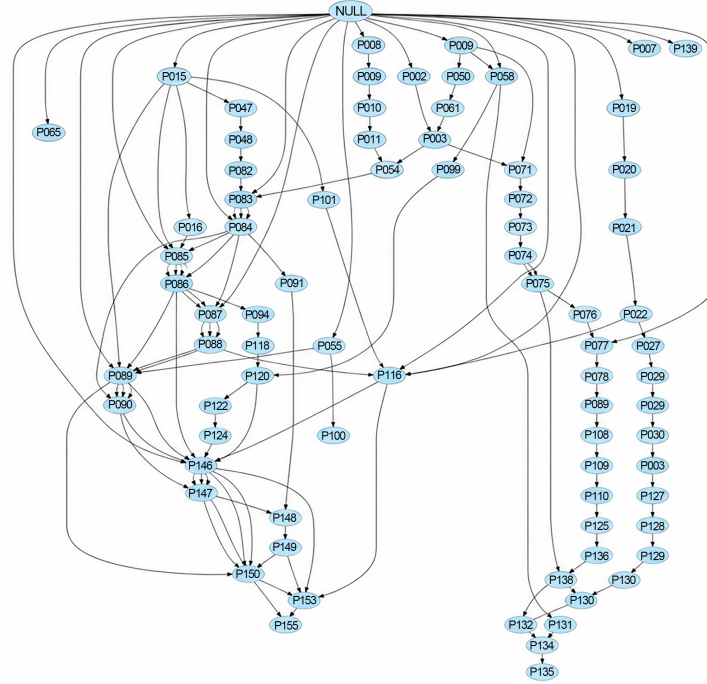


Figure 2. FP-Tree results from 30 Transactions.

Note: FP-Tree = Frequent Pattern Tree.

The resulting FP-Tree demonstrated that the most frequently occurring main path was the combination P015 → P04 → P010, with high frequency at each node. This result corroborated previous data that P015 and P04 had 100% support in the sample, while P010 had 86% support. In the branch section, nodes such as P013 and P017 appeared repeatedly on certain paths, indicating that these products often appeared as complements to core items though not as strongly as the main combinations. From a transactional knowledge discovery perspective, this FP-Tree served as a compressed representation of product co-occurrence patterns within transactions. The tree structure allowed the identification of “stable” purchase paths (e.g., combinations of household staples and daily necessities) as well as complementary patterns (products that tend to emerge when certain items are already in the basket). This approach aligns with other existing studies that adopted FP-Growth to map consumer purchasing patterns and provided insights for product placement and marketing strategies.

3.3 Frequent Pattern Growth Parameter Experiment and Selection of Optimal Support–Lift Values

In the full dataset (978 transactions and 157 products), the focus shifted from illustrating the FP-Tree structure to systematic experiments on FP-Growth parameters. The two major tested parameters were:

Minimum support (min. support): 0.03; 0.04; 0.05; 0.06; 0.07; 0.08; 0.09

Minimum lift (min. lift): 1.5; 2; 3

Table 3 shows an intuitive pattern emerged from sorting the experimental results based on the number of rules generated. The lower the min. support, the more rules were generated but this could lead to a certain number of irrelevant patterns (noise). The higher the min. support and min. lift, the fewer rules were generated but the remaining patterns were more specific.

The selection of a minimum support of 0.06 and a minimum lift of 2.0 was determined through a quantitative trade-off analysis between rule density and statistical significance. As recorded in Table 4, lower support values (e.g., 0.03) yielded a high volume of rules (up to 1,028), resulting in a high redundancy rate and many low-confidence

associations. Conversely, values above 0.07 significantly reduced rule coverage, capturing only the most obvious core patterns while missing subtle yet profitable cross-selling opportunities. The configuration of support = 0.06 and lift = 2.0 was selected because it yielded an optimal Average Confidence and a Rule Coverage of the active product catalog. This balance ensured that the rules were not only frequent enough to be reliable but also exhibited strong correlation (lift > 2), suggesting that the identified associations were twice as likely to co-occur under a random distribution. This systematic selection procedure provided a reproducible baseline for transactional knowledge discovery in this distribution context.

Table 3. Association rule results from each experiment

No.	Min. Support	Min. Lift	Rules Generated
1	0.03	1.5	1,028
2	0.04	1.5	418
3	0.05	1.5	302
4	0.06	1.5	152
5	0.07	1.5	70
6	0.08	1.5	34
7	0.09	1.5	22
8	0.03	2	966
9	0.04	2	388
10	0.05	2	280
11	0.06	2	140
12	0.07	2	64
13	0.08	2	30
14	0.09	2	22
15	0.03	3	790
16	0.04	3	344
17	0.05	3	254
18	0.06	3	124
19	0.07	3	54
20	0.08	3	26
21	0.09	3	22

Table 4. Experimental results sorted by the number of rules

No.	Min. Support	min. Lift	Rules Generated
1	0.03	1.5	1,028
8	0.03	2	966
15	0.03	3	790
2	0.04	1.5	418
9	0.04	2	388
16	0.04	3	344
3	0.05	1.5	302
10	0.05	2	280
17	0.05	3	254
4	0.06	1.5	152
11	0.06	2	140
18	0.06	3	124
5	0.07	1.5	70
12	0.07	2	64
19	0.07	3	54
6	0.08	1.5	34
13	0.08	2	30
20	0.08	3	26
7	0.09	1.5	22
14	0.09	2	22
21	0.09	3	22

3.4 Association Rules and Transactional Knowledge Generated

With the configuration $\text{min. support} = 0.06$ and $\text{min. lift} = 2$, FP-Growth generated a set of frequent item sets and association rules that represent the main purchasing patterns of customers. Although the entire list of rules was not presented in a large table here, the results section showed that core items such as P015, P04, and P010 often appeared as both antecedents and consequents, while other products such as P013 or P017, appeared as complementary items in certain combinations.

Substantively, high lift rules described practical knowledge such as (1) if customers purchase one or two core products (e.g., daily necessities), then the probability of them purchasing certain additional products will increase considerably; (2) there are groups of products that consistently “go together” in the same basket and they are turned into worthy candidates for bundling or cross-selling recommendations [16–19]. These observations accord with numerous studies which provided empirical evidence for the applicability of FP-Growth. The model could identify product combinations with strong correlations and enrich sales strategies across physical retail, e-commerce, and cafes and restaurants.

This article explicitly situated the pattern within the framework of transactional knowledge discovery, where association rules were not merely a list of rules but knowledge artifacts that were then structurally integrated into the database schema (product associations table) and used by the system module to support recommendation features of marketing strategies. To provide a more granular interpretation of these knowledge artifacts, a concrete example could be drawn from the generated rules. For instance, the association $\{P015, P04\} \rightarrow \{P010\}$ exhibited a lift value of 2.0. This indicated that a customer who purchased Product 015 (e.g., Indomie) and Product 04 (e.g., Kapal Api) was twice as likely as a random customer to also purchase Product 010 (e.g., Wardah).

From a managerial perspective, this rule translates into two practical decisions:

(1) Data-Driven Bundling: Management could design a “Consumer Essentials” package combining three high-correlation items to increase the average transaction value.

(2) Proactive Cross-Selling: The recommendation module in the system used this rule to alert distributors when a shop’s order is incomplete. This could ensure the preservation of the potential sales of P010.

Such specific applications illustrate how the FP-Growth output, beyond descriptive statistics, becomes a strategic tool for marketing optimization in the distribution sector.

3.5 Integration of Frequent Pattern Growth Results into the System and Its Methodological Implications

The deployment stage of the CRISP-DM framework was executed by operationalizing the association rules generated through FP-Growth within the business environment of UD XYZ. The rules discovered were stored as transactional knowledge and subsequently utilized to support data-driven marketing decisions, particularly product bundling and cross-selling strategies. The operationalization process was implemented through a web-based information system; however, the primary contribution of this study lies in the extraction and utilization of association rules rather than in the software implementation itself. The utilization of the association rules was exemplified by two practical business scenarios that represented the deployment of transactional knowledge within the organization:

(1) Used in the bundle management feature, where owners could be assisted in selecting statistically strong product combinations as promotional packages. The discovered association rules were utilized to support product bundling decisions by identifying statistically significant product combinations that frequently appeared together in historical transactions. These rules provide foreseeable insights for designing promotional packages and improving marketing strategies.

(2) Used in the product recommendation feature (cross-selling), which took customer transaction history and matched it with association rules in product associations to obtain additional product recommendations. The generated association rules were employed to support the implementation of cross-selling recommendations. By matching customer transaction histories with the discovered product associations, the system could recommend complementary products that exhibited strong co-purchasing relationships.

From a methodological perspective, this intriguing finding demonstrated that the CRISP-DM pipeline did not end at the modeling stage. Instead, the generated knowledge could be deployed and integrated into organizational decision-support processes. This approach aligns with growing studies that emphasized the operationalization of market basket analysis and association rule mining to substantiate practical business decision making rather than merely producing descriptive analytical results [20, 21].

4 Discussion

To clarify the contribution of this research, Table 5 contrasts the methodological and practical features of this study with those of representative prior work in association rule mining.

As shown in Table 5, while previous studies effectively identified patterns, this study extended the utility of FP-Growth by bridging the gap between algorithmic output and operational decision-making through systematic parameter optimization and software integration.

Table 5. Comparison of the proposed approach with existing studies on association rules

Feature Compared	Previous Studies [8, 11, 12, 21]	This Study (Proposed Framework)
Industry context	General retail/E-commerce	Middle-scale distribution sector
Process model	Algorithm-centric	Full CRISP-DM lifecycle
Parameter selection	Fixed/Arbitrary thresholds	Structured support-lift grid search
Knowledge output	Pattern listing/Visualization	Actionable transactional knowledge
Operational link	Static reports	Real-time system integration (API-based)

When compared with previous studies that applied the FP-Growth to analyze sales transactions, for example, in textile retail, cooperatives, wholesale stores, and cafes, most studies focused on identifying patterns and recommending marketing strategies such as cross-selling, product layout, or menu design [12, 21, 22]. This study shared the same objective but provided several unique methodological and practical contributions. First, each phase of business understanding, data understanding, data preparation, modeling, evaluation, and deployment was explained and executed consistently. This aligns with the current research trend that positioned CRISP-DM as the de facto standard for applied data mining projects, including those in data-driven marketing strategies [20, 23]. Second, instead of employing default values, this article conducted a small grid search over combinations of min. support and min. lift, monitored the number of generated rules, and selected the most balanced configuration for the context of distribution companies. This approach was rarely narrated in detail in the literature, even though the results could greatly affect the quality of the knowledge produced [24]. Apparently, the focus was on transactional knowledge discovery, not merely rule listing. Association rules were treated as knowledge artifacts that were integrated into the data schema (product associations table) and operationalized for two main business scenarios: bundle creation and cross-selling recommendations. This demonstrated how knowledge discovery from data mining was truly linked to functional modules in management information systems. Fourth, most FP-Growth studies shed light on modern retail, e-commerce, or medium- to large-sized businesses. This case study added the perspective of a distribution company in a region that previously used information systems only for transaction recording, not for analytics. Thus, this article enriched the literature on data-driven marketing in the context of emerging regions and traditional distribution businesses.

Overall speaking, the results of this study showed that the FP-Growth-based data mining method, orchestrated within the CRISP-DM framework, was capable of transforming previously passive transaction data into operational transactional knowledge. This included purchasing patterns (frequent patterns), association rules with high lift, and recommendations that could be directly used as the basis for bundling and cross-selling strategies. In distribution information systems, the role of FP-Growth as a knowledge discovery engine was strongly emphasized on the methodological pipeline.

5 Conclusions

This study enlightened the role of frequent pattern mining utilizing the FP-Growth algorithm within the CRISP-DM framework to extract valuable transactional knowledge from distribution companies' transaction data. From business understanding through deployment, the data mining process not only serves as a pattern-exploration tool, but also as an effective and structured analytical pipeline that transforms raw data into applicable knowledge. Experiments across various parameter settings showed that a combination of minimum support 0.06 and lift 2 yielded association rules that were most balanced between pattern quantity and quality, leading to relevant business interpretation.

The main findings of this study confirmed that core product patterns were consistent across nearly all transactions, whereas secondary association patterns formed cross-category purchasing relationships. These patterns demonstrated that FP-Growth could reveal latent structures in transactional data distributions that were previously invisible to conventional analysis. Thus, this study confirmed that data mining was the primary knowledge discovery engine for transactional data.

The chief contribution of this research lies in its methodological approach, which integrated the CRISP-DM-based full data-mining lifecycle with the FP-Growth parameter experiments and the deployment of analysis results into management information systems. Not only does it go beyond the discovery of frequent item sets and association rules, but it also demonstrates how these rules are systematically represented in the database and reused operationally, especially to support product bundling and cross-selling. This distinguishes the current research from most previous studies, which generally stopped at the pattern exploration stage without further considering the possibility of system integration.

In practical terms, the results in this study provided critical insights into the application of FP-Growth-based data mining in medium-scale distribution companies, in order to improve the quality of data-driven marketing decisions. Meanwhile, this research scientifically reinforced the position of frequent pattern mining as a key methodology in transactional knowledge discovery, especially when combined with standardized process pipelines such as CRISP-DM. Thus, the present study laid a solid foundation for progressing toward more adaptive, intelligent, and data-oriented marketing analytics systems in the future.

6 Limitations

Although this study successfully demonstrated the effectiveness of the FP-Growth in extracting transactional knowledge, several limitations must be acknowledged. First, the methodology relied solely on a single frequent pattern mining algorithm. While the FP-Growth is efficient for transactional compression, this research did not include a performance comparison with other established algorithms, such as Apriori or Eclat, which might offer different insights into computational speed or rule density in a distributional context. Second, the dataset was confined to a single distribution company within a specific province, which might constrain the generalizability of the findings to larger industrial scales or different geographical markets. Third, the evaluation of the association rules focused primarily on statistical metrics like support, confidence, and lift. A quantitative analysis of the direct financial impact, such as the actual increase in profit margins or stock turnover resulting from the implemented bundling strategies, were excluded from the scope of the current research. The bridging of these gaps could be prioritized in future research via conducting comparative algorithmic studies and validating the CRISP-DM pipeline across multiple datasets from various distribution sectors. Furthermore, integrating time-series analysis could explain seasonal purchasing patterns, while longitudinal studies could measure the long-term economic efficacy of data-driven marketing decisions within management information systems.

Author Contributions

Conceptualization, H.A. and L.H.; methodology, D.A.S.; software, D.A.S.; validation, H.A. and L.H.; formal analysis, H.A.; investigation, L.H.; resources, D.A.S.; data curation, D.A.S.; writing—original draft preparation, H.A.; writing—review and editing, L.H.; visualization, H.A.; supervision, L.H.; project administration, D.A.S.; funding acquisition, H.A. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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